Muthyala Koushik

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.17
Narayana Junior College	Class XII (Telangana State Board of Intermediate Education)	2024	96.50%
Narayana	Class X (Board of Secondary Education, Telangana State, India)	2022	8.80%

POSITIONS OF RESPONSIBILITY

SPICMACAY Coordinator | IIT Hyderabad

- Coordinated event planning, logistics, and volunteer operations for institute-level cultural programs.
- Collaborated with artists and institute administration to ensure seamless event execution.

SKILLS

Programming: C | Python | Verilog

Signal Processing: DSP | FFT | FIR Filters | Digital Filtering | Sampling & Reconstruction

Hardware: FPGA Design | Embedded Systems | Analog Circuit Design

Tools: MATLAB | Quartus Prime Lite | LTspice | LaTeX | Git | GitHub | Linux | Arduino IDE

RELEVANT COURSES

Core Courses: Signal Processing | Analog Electronics | Digital Systems | Semiconductor Device Fundamentals | Control Systems

Lab Courses: DSP Lab | Electronic Devices & Circuits Lab | Digital Systems Lab | Electrical Circuits Lab

Programming & Mathematics: Introduction to Programming | Machine Learning | Probability | Random Processes | Matrix Theory | Differential Equations & Transform Techniques

PROJECTS

FPGA-Based Pipelined Radix-2 FFT & FIR Accelerator | Verilog, Intel Quartus Prime, MATLAB

- Designed and implemented a **pipelined Radix-2 FFT** and **fixed-point FIR filter** in Verilog for real-time DSP applications.
- Developed a **pipelined RTL architecture** with optimized control and datapath logic to maximize throughput and efficiency.
- Optimized logic utilization while balancing **pipeline latency**, operating frequency, and **FPGA resource utilization**.
- Synthesized and analyzed the design using **Intel Quartus Prime**, meeting functional and timing requirements.
- Verified functionality against **MATLAB** reference models and validated the design on **FPGA hardware**.

Digital Lock-In Amplifier | Altium Designer, STM32H743

- Designed a digital **lock-in amplifier** using the **STM32H743** for high-sensitivity signal recovery applications.
- Developed an **analog front end** with instrumentation amplification, offset conditioning, and anti-aliasing filters.
- Implemented **I/Q synchronous demodulation** and digital signal processing for accurate amplitude and phase extraction.

- Integrated analog and digital subsystems to ensure reliable **signal acquisition**, conditioning, and system performance.
- Optimized component selection and circuit integration to minimize noise while improving **signal stability** and accuracy.

Microphone Amplifier with Class-AB Driver | LTspice, Oscilloscope

- Designed and implemented a **Class-AB push-pull power amplifier** with a low-noise microphone preamplifier.
- Developed transistor biasing and output-stage circuitry to reduce **crossover distortion**.
- Characterized amplifier performance by measuring **THD**, **PSRR**, voltage gain, bandwidth, and output swing.
- Performed **FFT-based** measurements to evaluate frequency response, harmonic distortion, and output power.
- Verified speaker-driving capability through experimental validation of audio quality and bias stability.

Sampling Theorem Demonstration using SAR ADC & R-2R DAC | Arduino, C, Oscilloscope

- Designed and implemented a **4-bit SAR ADC** and **R-2R ladder DAC** for analog signal sampling and reconstruction.
- Developed **SAR control logic** with comparator-based successive approximation and ring-counter-driven bit selection.
- Validated the **Nyquist Sampling Theorem** under sub-Nyquist, Nyquist, and oversampling conditions.
- Characterized converter performance by measuring **DNL**, **INL**, resolution, and conversion time.
- Verified reconstruction fidelity through oscilloscope-based analysis of **aliasing** and reconstructed analog waveforms.

Discrete Fourier Transform & Frequency-Domain Signal Analysis | Python, NumPy, MATLAB

- Implemented **DFT**, **IDFT**, and matrix-based DFT algorithms from first principles in Python.
- Verified fundamental **DFT properties**, including linearity, circular shift, Parseval's theorem, and convolution.
- Analyzed **magnitude** and **phase spectra** of discrete-time signals using frequency-domain visualization.
- Applied **frequency-domain analysis** for spectral analysis and filtering of sinusoidal and composite signals.
- Validated custom implementations against **NumPy FFT/IFFT** to verify correctness and computational performance.

EXTRACURRICULAR

National Service Scheme (NSS) Volunteer | Interests: Movies, Anime, Cricket, and Badminton.